Section A

1	**(a)(i)**	Period = 0.80 s Frequency = 1/T = 1/0.80 = 1.25 Hz	**C1** **A1**
	(a)(ii)	Magnitude of induced emf is proportional to the rate of change of flux linkage	**A1**
	(a)(iii)	1. As magnet oscillates in the coil, the flux linkage with the coil changes and emf is induced in the coil, from Faraday's Law. 2. This causes an induced current to flow when the switch is closed. 3. The oscillating system loses energy as there is heat loss in the resistor. 4. Since energy of the system is proportional to the square of amplitude, amplitude decreases. OR 3. From Lenz Law, the flow of induced current produces a magnetic field which results in a force opposing the motion of the magnet. [B1] 4. The maximum resultant force on the magnet decreases and since amplitude is proportional to the maximum resultant force, amplitude decreases. [B1]	**B1** **B1** **B1** **B1** **OR** **B1** **B1**
	(b)(i)	0.50 Hz	**A1**
	(b)(ii)	Amplitude increases then decreases Amplitude is maximum at 1.25 Hz when resonance occurs. (Amplitude of oscillation increases as freq increases from 0.5 Hz to 1.25 Hz. At 1.25 Hz, resonance occurs. As frequency increases beyond 1.25 Hz to 5.0 Hz, the maximum amplitude decreases)	**B1** **B1**

2	(a)(i)	Single slit: $\sin\theta = \lambda/b$	C1